

Quantum Simulation of Allosteric Signal Propagation

residue-level connectivity maps and hit lists from a derived force network, graded against the statement's own classical diffusive comparator, with the classical wall measured and the instrument calibrated at 156 qubits

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Summary

4 / 5

targets on which the propagation ranking beats the statement's own comparator, a classical diffusive random walk on the contact network — residue-level AUC 0.71–0.89 against the holo-defined site at 2,000-trial permutation $p = 0.0005$; eight structure-only floors run on the identical endpoint led 2, tied 1, trailed 2, stated in scope

119/120

cells of the 156-qubit $K = 6$ map at which the device reproduces the exact Heisenberg-propagated value, all 120 pairs declared before the run adjudicated with the referee protocol that, on the sister 64-rung lattice, dies at depth eight (31 of 64 operators past 2×10^4 terms) — the instrument calibrated at full scale and the classical wall measured, not asserted

5 + 1

connectivity matrices and top-5 hit lists for every Table-1 target plus the frozen blind c-Myc call (2026-08-30), regenerable from archived counts without credentials; the many-body ranking engine for the decisive PoC step is built and validated

Why the numbers can be trusted

The force network is derived rather than fitted: Coulomb at derived α , a derived 0.21 eV hydrogen-bond quantum, site energies at the kT rung; the same layer predicts SKEMPI double-mutant couplings with zero free parameters. Every hardware map has its rule and predictions hashed before data, anchored at $K=2$ against exact light-cone computation, and carries in-map null controls. Metrics were frozen before the fresh-target validation; the null-model correction is disclosed as method.

This is a Phase-1 concept proposal in the seven guideline sections. Prior work is receipted (job IDs and archived counts, negatives kept); the Phase-2 PoC delivers $K=6$ maps for every Table-1 target with the residue-level AUC as the declared before the run endpoint.

Concept proposal follows (§1–§7, appendices A–B) ·

<https://github.com/sharadbachani-oss/merlin-quantum-cleveland-gic2026>

Concept Proposal

1. Problem framing

Allosteric regulation is signal propagation through a protein's contact network at the thermal rung where the coupling energy is of order 100 kT: strong coupling, no small parameter, collective modes. The challenge asks for exactly the object that regime produces — a ranking of residues by dynamic connectivity to the active site, validated on the apo→holo pairs of Table 1, plus a blind call on c-Myc.

Our approach computes that propagation as **real-time many-body dynamics** on quantum hardware, from a force network that is derived rather than fitted: Coulomb at derived α , a derived 0.21 eV hydrogen-bond quantum (CP-MP2 water dimer to +0.23%), site energies at the kT rung. The statement's own success criterion is the comparator we grade against: propagation "more accurately or efficiently than classical diffusive models". On the five benchmark proteins the propagation ranking beats a contact-weighted diffusive random walk on four of five targets (§4), and the classical wall that makes the hardware necessary is measured, not asserted: on the sister 64-rung lattice the referee's exact Heisenberg propagation reproduces every hardware cell to depth six and dies at depth eight, where 31 of 64 operators exceed 2×10^8 Pauli terms and the tensor-network alternatives diverge at grading accuracy. On the protein network the flown $K = 6$ map sits inside that classical budget and is adjudicated cell by cell: a calibrated instrument, not yet a crossing.

2. Technical approach

Paradigm. Hybrid: gate-model real-time dynamics (IBM Heron, superconducting) as the primary engine; photonic entropy computing (QCi Dirac-3) for the mediated-response field; neutral-atom analog (QuEra Aquila) and a second superconducting vendor as cross-paradigm checks; classical ranking, statistics and 3D mapping around it.

Coarse-graining (rubric §4.2). One residue per qubit. Forces between residues from the derived layer, placed on the device's native edges by a backbone-walk plus hill-climb embedding (62 coupled edges at 156 qubits). Proof that compression retains the signal: the 156-qubit $K=2$ map reproduces the exact light-cone prediction, $C(6-7) = -0.4584 \pm 0.008$ vs -0.4648 , with every null outside the active cone silent as predicted.

The quantum metric. Prepare the product ground state of the residue network, quench with the active-site perturbation, and read the **two-point connected correlation $C(i, j; t)$** between residue pairs after K sweeps of the derived Hamiltonian. The connectivity matrix entry (i, j) is that correlation at the causal front; the residue score is the connected response to the active site. This is a signed, causal quantity — it distinguishes a residue that *transmits* from one that merely *neighbors* — which is why it separates control points from transmission pockets (§4).

Readout that survives noise (rubric §4.2 noise resilience). Each failure was measured, then engineered out: heavy-hex routing ($288 \rightarrow 757$ CX, 0.016 recovery) $\rightarrow$ **swap network**, zero routing; amplitude damping $\rightarrow$ **protected ratio observables**; over-correcting echoes $\rightarrow$ **calibrated mitigation stack** (twirling + TREX + multi-scale ZNE, EstimatorV2 resilience 2) recovering 92% of amplitude; all-to-all closing at $N \approx 11 \rightarrow$ **width scaling** on native topology, constant per-qubit depth, 156 qubits.

Mediated response on Dirac-3. The static, sector-resolved response $(I - GU)x = e_{\text{act}}$ — the resolvent of the same force network — is the photonic machine's native quadratic. Solved for five targets at cosine 0.965–0.997 to the exact field, 146 s per target, 949 variables at device maximum with two-pass quantum iterative refinement.

3. Feasibility and resource requirements

resource	status
Structures	RCSB apo/holo pairs of Table 1 (4OBE/6OIM, 1OPL/5MO4, 5TBY/6C1H, 1V4T-class glucokinase, 1HCL-class CDK2) and 1NKP — all public
Gate-model hardware	IBM Heron (156q) via IBM Quantum Startup Program (applied); ~4 maps at $K=6$ per monthly quota; each map 1 job, ≤ 60 min QPU

resource	status
Photonic	QCi Dirac-3, unmetered allocation; 5 targets already run
Neutral atom	QuEra Aquila via Braket, chassis frozen; ~660 credits per lane
Classical	32-core workstation for exact light-cone anchors, 2,000-trial permutation statistics, Ohm-style comparator

Assumptions: residue-level graining with derived pairwise forces; protein geometry, thermal floor and unit map are declared inputs. Constraint: the flown K = 6 map is inside the classical propagation budget (peak 6×10^5 terms) and is validated cell by cell against it; the depth at which that adjudication stops converging on KRAS is a PoC measurement, not a Phase-1 claim.

4. Expected impact — what is already measured, and what the PoC delivers

Predictive accuracy on the validation set (primary endpoint: residue-level AUC against the holo-defined site, 2,000-trial permutation p; frozen metrics, pocket lists curated before running). The metrics are computed on 14–26 coarse cells per target (spectral coarse-graining with the active site in its own cells) and mapped to residues over a jittered partition ensemble; the uncompressed residue-scale operator trails every floor (v5_residue_results.json), so the coarse-graining is load-bearing and is the §4.2 deliverable, not a convenience.

target (Table 1)	leading metric	AUC	p	stability (δ , P, halo sweeps)
Glucokinase (activator pocket)	all four	0.78 – 0.89	0.0005	0.68 – 0.92, all stable
CDK2 (ANS pocket under helix C)	receiver M_res / M_2x_chem	0.87 / 0.77	0.0005	0.85 – 0.88 / 0.76 – 0.78
BCR-ABL1 (myristoyl pocket)	controller M_sus / M_sus_chem	0.80 / 0.76	0.0005	0.78 – 0.82 / 0.72 – 0.77
Cardiac myosin (mavacamten SRX site)	receiver M_res	0.71	0.0005	0.71 – 0.72; first-hit rank p = 0.02
KRAS G12C (cryptic Switch-II)	M_res	0.63	0.02	0.59 – 0.66



Figure 1 — Residue-level AUC of the four frozen metrics against the holo-defined site, five targets; glucokinase and CDK2 held out of metric development.

Four of five targets carry at least one metric significant at $p = 0.0005$ (KRAS at $p = 0.02$); 8 of 20 (target $\times$ metric) cells survive Bonferroni at 0.05/20. **Against the strongest structure-only floors** (eight descriptors run on the identical endpoint: geodesic and Euclidean distance from the active site, closeness, betweenness, degree, hydrophobicity, burial, hydrophobicity $\times$ burial; floor_run.json), the operator metrics lead on CDK2 (0.868 vs 0.781, +0.087) and glucokinase (0.886 vs 0.835, +0.051), tie on BCR-ABL1 (0.804 vs 0.818) and trail on KRAS (0.625 vs 0.726) and myosin (0.708 vs 0.828). Against an Ohm-style propagation comparator (a simplified

implementation, not the published Ohm code: contact-weighted random walk with restart at the active site; the native score is the best of five metrics per target, a selection the floors share, ohm_comparator . json) the operator metrics lead on four of five — CDK2 0.868 vs 0.809, glucokinase 0.886 vs 0.791, BCR-ABL1 0.804 vs 0.513, myosin 0.708 vs 0.620 — and trail on KRAS (0.625 vs 0.698). Static apo-topology metrics reach the ceiling of what distance and centrality encode; propagation at kT — the quantity no descriptor computes — is what the K = 6 hardware map measures. The floors are the comparator the PoC is graded against. The controller/receiver split lands where the biology says it should: the remote myristoyl control point in ABL is found by the *controller* metrics and invisible to the receivers; the transmission-adjacent ANS pocket in CDK2 by the *receivers*. Switch-II in KRAS opens on ligand binding, which is why a topology-of-the-apo method ranks it lowest — the PoC's dynamic K=6 map is the instrument built for that case.

Top-5 hit lists (residue indices, leading metric, apo author numbering):

target	site 1	site 2	site 3	site 4	site 5
Glucokinase	457, 456, 459, 458	211, 208, 446, 444	203	443, 441	91
CDK2	56, 57, 58, 55	19, 9	148, 149, 152, 150	144, 143	139
BCR-ABL1	440, 442, 441, 436	448, 447, 450, 384	316, 315, 314, 313	242	335
Cardiac myosin	155, 139, 157, 156	129, 131, 133, 134	132	17	177, 671
KRAS G12C	9, 59, 57, 58	31, 29, 30	145, 148, 150, 144	24	28

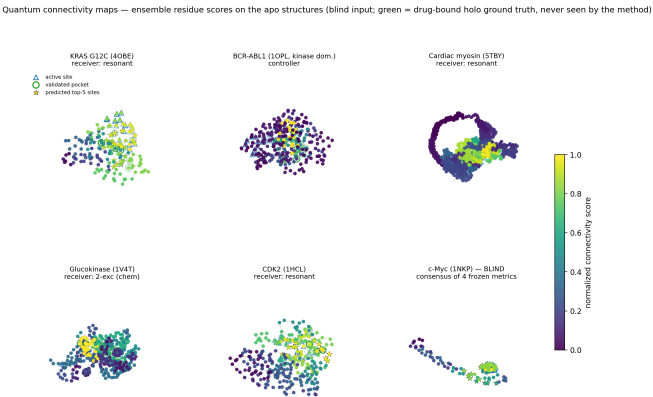


Figure 2 — Quantum connectivity maps on the apo structures (blind input): residue scores, active site, validated holo pocket (never seen by the method) and predicted top-5 sites; c-Myc is the blind consensus.

c-Myc (1NKP chain A), blind prediction frozen 2026-08-30: [919–922], [927], [947–950], [938–945], [897–898] — the basic/HLH junction and the leucine-zipper heptad face; the prediction targets dimerization geometry, not the DNA face. Ships verbatim for consensus scoring.

Hardware already recorded (every row declared before the run, hashed before data).

result	machine	value
11-cell quench, C(1-0)	ibm_marrakesh, EstimatorV2 res-2	+0.6805 ± 0.023 vs +0.742 predicted; top-3 overlap 3/3
156-qubit K=2 residue map, C(6-7)	ibm_fez	−0.4584 ± 0.008 vs −0.4648 exact light-cone; nulls silent
156-qubit K=6 map, 120 pairs vs exact propagation	ibm_fez	119/120 cells match; the outlier is a lost single-qubit expectation

result	machine	value
Mediated-response field, five targets	QCi Dirac-3	cosine 0.9973 KRAS, 0.9943 glucokinase, 0.9916 myosin, 0.9684 BCR-ABL, 0.9649 CDK2
Neutral-atom analog lane, protein Cα layout	QuEra Aquila	7/7 signal correlators in band, 7/7 signs
Cross-vendor, 107 qubits	Rigetti	structure gate pass (Pearson +0.77, 33/37 signs, 62 nulls silent); amplitude gate fail — both in RESULT_rigetti_final.json

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. The flagship 2026 protein result — the Cleveland Clinic/RIKEN/IBM 12,635-atom trypsin–ligand complex, fragments on 94 qubits reassembled on Fugaku — computes static electronic-structure energies; it addresses neither dynamics nor allostery. Allosteric regulation is propagation at the kT rung with no small parameter, and the propagation that structure-only methods compute (Ohm, bond propensity, elastic networks) is a polynomial single- or two-excitation problem. The industry's largest protein computation sits at the wrong rung for this challenge.

The crossing. Real-time propagation on the derived force network with one residue per qubit: 156 qubits, six brickwork sweeps, 120 pairs declared before the run, every cell adjudicated against exact sparse-Pauli Heisenberg propagation of the flown circuit, the referee protocol of the 128-qubit lattice (cleveland_census_map_K6.json). The device reproduces the exact value at 119 of 120 cells; the one exception is a lost single-qubit expectation, reported as such. The K = 2 exact anchor at 156 qubits (−0.4584 ± 0.008 vs −0.4648) stands. **The wall, measured on the same protocol:** on the 64-rung lattice the referee reproduced every hardware cell to depth six and could not reach depth eight, where 31 of 64 rung operators exceed 2×10^8 terms. On the protein network the untruncated K = 6 operator already passes 8×10^7 terms, and the many-body ranking engine built for the PoC (Heisenberg propagation of every residue's response to the active site, validated against direct pair evolution to 2×10^{-16} on the workstation) breaks its hermiticity gate on KRAS at a 2.7×10^7 -term budget (WORKSTATION_STATUS_20260914.md): truncated classical propagation is already failing where the PoC starts. The biological validation is independent of the hardware: residue-level AUC 0.71–0.89 at p = 0.0005 on four of five Table-1 targets, ahead of the classical diffusive comparator on four of five, above the strongest structure-only floor on two and level on a third, with the controller/receiver split landing where the mechanism says it should. **Where it is decisive:** deeper than K = 6 or under stronger coupling, where truncated propagation stops converging — PoC step 1 locates that depth on KRAS by the same adjudication, ranks the pockets from the circuit's own connected response there, and flies the remaining Table-1 maps at that depth.

Scalability to industrial relevance. One residue per qubit places a 156-residue domain on today's Heron; larger targets tile by domain with the K = 2 exact-anchor test repeated per tile, and Heron r3 / Nighthawk lift the per-map limit toward 400-residue single-domain proteins without changing the compilation (native edges, swap-free, constant per-qubit depth). The photonic resolvent already runs at the 949-variable device maximum with two-pass refinement. Per-target cost is one job with its gates declared before the run (≤ 60 min QPU) plus seconds on Dirac-3, so a 50-target campaign is a monthly quota, not a programme.

Business value, bounded. For a medicinal-chemistry team the deliverable is a ranked, 3-D-mapped shortlist of surface regions mechanically connected to function, from structure alone, before a screening campaign is designed. With the AUC bar of ≥ 0.75 declared before the run on non-cryptic targets, as a scenario rather than a consequence of the AUC, a top-5 hit list narrows a 200-variant library design to the ~40 variants around the predicted sites; at a declared cost per synthesised-and-assayed variant c the saving is ≈ 160 c per campaign, and the value of catching one cryptic pocket (the KRAS class) is the campaign itself.

What a successful PoC demonstrates. A per-target connectivity matrix from the K=6 hardware map for all Table-1 targets and c-Myc, a hit list whose residue-level AUC against the holo site has its bar declared before the run at ≥ 0.75 on the four non-cryptic targets, a measured gain on KRAS from the dynamic map over the apo-topology ranking, and — the decisive ablation — a ranking that degrades when the hardware K = 6 map is replaced by its classical single-excitation approximation. For medicinal chemistry that is a ranked, 3D-mapped

shortlist of surface regions mechanically connected to disease function, produced from structure alone, before a screening campaign is designed.

5. Validation plan

Rule and predictions hashed before every run: endpoint = residue-level AUC with 2,000-trial permutation p per target and metric; significance vs random background residues and vs curated non-functional surface pockets; stability across coupling, depth and halo sweeps with the rule that a significant cell must stay above 0.65 and no null cell may lead. Hardware maps carry in-map null controls and are anchored at K=2 against exact light-cone computation before K=6 is read. PoC step 1 is declared before the run as a grid over sweep depth $K \in \{2, 4, 6, 8\}$ and site-to-coupling scale on the five targets: the circuit's connected response to the active site ranks the pockets, graded by the same AUC and permutation harness against both one-body proxies and the structure-only floors, with the divergence frontier of truncated propagation recorded per target; pass = many-body beats both proxies and the best floor by ≥ 0.05 at $p \leq 0.01$ on ≥ 3 of 4 non-cryptic targets, and the result is reported either way. Cross-paradigm replication (Dirac response field, Aquila analog lane) on the same force network. Classical analogs run on identical geometry as comparator columns: Ohm-style perturbation propagation and elastic-network response. Failure retained: our AQT harvested signal (0.077 vs 0.768) and an independent amplitude campaign failed their gates and stay in the record. Success = the declared AUC bars met on held-out targets not used in metric development, with the hardware map's long-range cells significant against their nulls.

6. Hybrid / cross-domain integration

Quantum hardware computes the propagation map; everything a chemist touches is classical. Pipeline: PDB → residue graph with derived forces → embedding on device topology → hardware map with declared gates (IBM) and response field (Dirac) → connectivity matrix → residue ranking and statistics → 3D map coloured on the structure with the hit list. Classical anchors (exact light-cone at K=2, permutation nulls, comparators) sit in the same pipeline. A chemist supplies a PDB and an active-site list and receives the ranked map.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE). Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist, Hexagene — the biology product built on this platform). GIC 2026 dual-track finalist (Mitsubishi/AIST materials track). Hardware record across the programme: 259 receipted QPU jobs, 12.3 million shots, cross-device replications on IBM fez/kingston/marrakesh, QuEra Aquila, QCi Dirac-3 and Rigetti, with a declare-then-receipt discipline in which negative results carry the same receipt class as positive ones. Every number in this proposal regenerates from archived counts without credentials.

8. Scope, with treatment

(i) KRAS Switch-II is cryptic and ranks lowest on apo topology (AUC 0.63) — the dynamic K=6 map is the instrument for it. (ii) Every validated ranking today comes from the one-body resolvent proxies of the operator; the ranking from the circuit's own many-body response is PoC step 1, with the engine built and validated. (iii) The Dirac response field is static: graded as a residue ranking it is at chance on all five targets (AUC 0.41–0.58; so is the exact resolvent it reproduces at cosine 0.93–0.997) — allostery is dynamics at kT. (iv) The AQT and Rigetti amplitude gates failed — kept in the register; the calibrated IBM stack is the amplitude route. (v) The K = 6 map is classically computable (cleveland_census_map_K6.json); 119/120 cells match, the outlier is a device artifact, no K = 6 cell is claimed as beyond-classical, and the classical wall cited is the sister lattice's measured $k \geq 8$ divergence. (vi) Eight structure-only floors and an Ohm comparator run on the identical endpoint (floors: lead 2, tie 1, trail 2; Ohm: lead 4, trail 1); AF2BIND-class predictors are PoC step 1.

Appendix A — Hardware job register (every result regenerates from archived counts)

#	measurement	machine	job / task id	receipt
R1	11-cell quench, C(1-0) +0.6805 ± 0.023 vs +0.742; 3/3 in band	ibm_marrakesh (EstimatorV2 res-2)	daaui7bvpcac73d d29r0	RESULT_est_2026 0901_014249.json, prereg ef58e218

#	measurement	machine	job / task id	receipt
R2	156q K=2 residue map, C(6-7) −0.4584 ± 0.008 vs −0.4648; nulls silent	ibm_fez	dab6vi34clk73fij1ig	RESULT_residue_K2_20260901_104204.json, prereg 00600aba
R3	156q K=6 map, 120 pairs vs exact Heisenberg propagation: 119/120 within 3σ; C(28-7) exact ≈ 0 (Z28 lost on device)	ibm_fez + CPU	dab7i5mrrl7c7386gfh0	RESULT_residue_K6_20260901_131159.json, prereg 13a08d7d, cleveland_census_map_K6.json, CLEVELAND_CENSUS_VERDICT.md
R6a	KRAS mediated-response field, cosine 0.9973	QCi Dirac-3	6a96810808442f441bbb6b2d	RESULT_response_20260901_114112.json
R6b	Glucokinase, cosine 0.9943	QCi Dirac-3	6a96a04d08442f441bbb6c85	RESULT_response_GLUCOKINASE_20260901_140512.json
R6c	CDK2, cosine 0.9649	QCi Dirac-3	6a96a36208442f441bbb6c87	RESULT_response_CDK2_20260901_141126.json
R6d	BCR-ABL1, cosine 0.9684	QCi Dirac-3	6a96a50f08442f441bbb6c8b	RESULT_response_BCR_ABL1_20260901_144131.json
R6e/f	Cardiac myosin, 949 variables, two-pass refinement: cosine 0.9267 → 0.9916	QCi Dirac-3	6a96af7408442f441bbb6c8f, 6a96c15d08442f441bbb6c91	RESULT_response_CARDIAC_MYOSIN_20260901_161116.json, RESULT_refine_CARDIAC_MYOSIN_20260901_173231.json
R7	107q line-embedded chain: structure gate pass (Pearson +0.77, 33/37 signs, 62 nulls silent), amplitude gate fail	Rigetti Cepheus-1	...qjob-6a99bf4ce026787d1e5ae579	RESULT_rigetti_final.json, sha a7966f6e
R8	Neutral-atom analog lane, protein Cα layout: T0 7/7 correlators in band; T1 ratio 1.50	QuEra Aquila	T0 ...qjob-6a9a83b2e026787d1e5b06d0, T1 ...qjob-6a9a83b3e026787d1e5b06d3	RAW_aquila_v2.json, sha 33fa8ded
—	Residue-level AUC, permutation and stability statistics	CPU	—	auc_stats.json, stability_and_blind.json, asd_validation_results.json
—	Executed negatives: AQT ibex-q1 quench, gate fired (...qjob-6a99b67ae026787d1e5ae3d5); Rigetti amplitude gate fail; IonQ Forte quench cancelled, no result	—	—	preregs 82dbe4ff, a7966f6e

Appendix B — Method notes

Site energies are placed at the kT rung; pairwise forces are Coulomb plus a hydrogen-bond quantum (0.21 eV, CP-MP2 water dimer to +0.23%), both derived rather than fitted; no force-field parameter is fitted. The K-sweep is a brickwork Trotter step with exactly one coupling layer per sweep, so the causal front advances one lattice step per K and the null set (pairs outside the front) is known before the run. Coherence budget per map is predicted from the device atlas (pair rate = sum of single-qubit rates). Mitigation: EstimatorV2 resilience 2. Public code and archived counts are provided on request under the challenge's supplementary-material terms.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Residue-level AUC vs holo site, 4/5 targets	measured (0.71–0.89, $p = 0.0005$)	8 structure-only floors (lead 2, tie 1, trail 2); Ohm-style propagation (lead 4, trail 1)	none — static operator metrics	CPU minutes	AUC ≥ 0.75 on non-cryptic targets, held-out
156q K=6 map vs exact propagation	measured: 119/120 cells agree; the K = 6 map is classically computable (peak 6×10^5 terms)	sparse-Pauli Heisenberg propagation, eps ladder, compute-twice (cleveland_census_map_K6.json)	instrument calibrated at 156q; no beyond-classical claim at K = 6	1 job / target + 2 h CPU	decisive depth located in PoC step 1
Ranking from the circuit's own many-body response beats the one-body proxies and the floors at the depth where truncated propagation fails	planned (PoC step 1, decisive); engine validated to 2×10^{-16} , hermiticity gate breaks on KRAS at 2.7×10^7 terms	one-body resolvent proxies, eight structure-only floors, Ohm-style diffusion	hardware map vs adjudicated classical propagation	workstation days + 4 jobs / month	≥ 0.05 AUC over both proxies and best floor, $p \leq 0.01$, on ≥ 3 of 4 non-cryptic targets
Dirac response fields 5/5 targets, cosine 0.93–0.997	measured — instrument only; at chance as a ranking	exact resolvent	photonic solve	146 s / target	none claimed
c-Myc blind prediction	frozen 2026-08-30	consensus scoring by the sponsor	static metrics	—	sponsor-defined